

Guided Notes: Making k-Nearest Neighbors Predictions

These notes focus on what k-nearest neighbors (k-NN) does at prediction time. The central procedure is simple: measure distance from a query point to the training data, keep the k closest observations, and aggregate their outcomes.

Learning goals

By the end of these notes, you should be able to:

1. calculate distances from a query point to training observations;
2. identify the k nearest neighbors;
3. make unweighted k-NN predictions for classification and regression;
4. make distance-weighted k-NN predictions;
5. explain how the distance metric, feature scaling, and k affect a prediction; and
6. handle ties and zero-distance observations deliberately.

1. The k-NN prediction recipe

Let the training data be

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\},$$

where $x_i \in \mathbb{R}^p$ is the feature vector for observation i , and y_i is its outcome. For a new query point x_0 , k-NN follows this procedure:

1. Choose a distance function $d(x_0, x_i)$.
2. Compute the distance from x_0 to every training point.
3. Keep the k training points with the smallest distances; call this set $N_k(x_0)$.
4. Aggregate their outcomes to produce $\hat{y}(x_0)$.

Fill in the language

Quantity	Meaning
x_0	
x_i	
$d(x_0, x_i)$	
$N_k(x_0)$	
k	

Why is a distance function part of the k-NN model, rather than merely an implementation detail?

2. Distances determine the neighborhood

For feature vectors $x = (x_1, \dots, x_p)$ and $z = (z_1, \dots, z_p)$, common choices include

Distance	Formula	When it may be natural
Euclidean (L_2)	$d_2(x, z) = \sqrt{\sum_{j=1}^p (x_j - z_j)^2}$	Numeric features with comparable scales
Manhattan (L_1)	$d_1(x, z) = \sum_{j=1}^p x_j - z_j $	Axis-aligned differences; less emphasis on large gaps
Hamming	$d_H(x, z) = \sum_j \mathbf{1}\{x_j \neq z_j\}$	Binary, categorical, or fixed-length discrete features

Mahalanobis	$d_M(x, z) = \sqrt{(x - z)^T \Sigma^{-1} (x - z)}$	Correlated numeric features with a reliable covariance estimate
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A quick distance calculation

Let $x_0 = (2, 1)$ and $x_i = (5, 5)$. Compute both distances:

$d_2(x_0, x_i) =$ _____, $d_1(x_0, x_i) =$ _____.

Will Euclidean and Manhattan distance always rank two candidate neighbors in the same order? Explain.

Scale is part of the distance calculation

Suppose one feature is age (roughly 18–80) and another is annual income (roughly \$20,000–\$200,000). Before applying L_1 or L_2 distance, what should you consider doing? Why?

3. Constructing the neighborhood

For a query x_0 , the k -nearest neighborhood contains the k points with the smallest distances:

$$N_k(x_0) = \{(x_i, y_i) : x_i \text{ is among the } k \text{ closest training points to } x_0\}.$$

If $x_i \in N_k(x_0)$ and $x_j \notin N_k(x_0)$, then

$$d(x_0, x_i) \leq d(x_0, x_j).$$

Prediction worksheet data

Use Euclidean distance throughout this example. A query point is

$$x_0 = (2, 1).$$

The training data contain a class label and a numeric response. We will use the same neighbors for classification and regression.

ID	Feature vector x_i	Class label	Numeric response y_i	$d_2(x_0, x_i)$	Rank
A	(2, 2)	Red	8		
B	(3, 1)	Blue	10		
C	(0, 1)	Red	4		
D	(2, 4)	Blue	14		

1. Compute each distance and rank the observations from nearest to farthest.
2. Circle or list $N_3(x_0)$.
3. What is the radius of the smallest Euclidean ball around x_0 that contains the three selected neighbors?

Boundary ties

Suppose the second- and third-smallest distances are equal, but you have chosen $k = 2$. What decision must an implementation make?

4. Unweighted k-NN predictions

Classification: majority vote

For a class label c , count how many of the neighbors have that label:

$$\hat{y}(x_0) = \arg \max_c \sum_{i \in N_k(x_0)} \mathbf{1}\{y_i = c\}.$$

Using the $k = 3$ neighborhood from Section 3, complete the vote table.

Class	Number of votes	Weighted score (leave blank for now)
Red		
Blue		

$$\hat{y}_{\text{class}}(x_0) = \underline{\hspace{2cm}}.$$

If the vote is tied, name one reasonable tie-breaking rule.

Regression: uniform average

For a numeric outcome, unweighted k-NN uses the arithmetic mean:

$$\hat{y}_{\text{reg}}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} y_i.$$

Using the same three neighbors, show the calculation:

$$\hat{y}_{\text{reg}}(x_0) = \underline{\hspace{2cm}}.$$

How would the prediction change if $k = 1$? What possible benefit and drawback come with that choice?

5. Distance-weighted k-NN predictions

Uniform k-NN gives each selected neighbor the same influence. A distance-weighted version gives closer neighbors more influence. One common choice is inverse-distance weighting:

$$w_i = \frac{1}{d(x_0, x_i)}.$$

For regression, the weighted prediction is

$$\hat{y}_{\text{wreg}}(x_0) = \frac{\sum_{i \in N_k(x_0)} w_i y_i}{\sum_{i \in N_k(x_0)} w_i}.$$

For classification, one weighted-vote version is

7. A prediction checklist

Before reporting a k-NN prediction, verify the following.

- ☐ The query point and training features use the same representation and preprocessing.
- ☐ The distance metric is appropriate for the feature types and scales.
- ☐ Distances have been computed consistently.
- ☐ Exactly k neighbors have been selected, with a documented rule for boundary ties.
- ☐ The aggregation rule matches the outcome type: vote for classification, average for regression.
- ☐ A zero-distance rule has been specified when using inverse-distance weights.

Exit questions

1. Can changing the distance metric change a k-NN prediction even when k is fixed? Why?
2. In weighted k-NN regression, why does the denominator $\sum_{i \in N_k(x_0)} w_i$ appear?
3. A classifier has neighbors with labels A, A, B, B for $k = 4$. Give two possible tie-breaking rules.
4. Why can a feature measured on a very large numeric scale dominate Euclidean distance?